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Human-Centered Artificial Intelligence: A Field Experiment

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Abstract. Humans and artificial intelligence (AI) algorithms increasingly interact on unstructured managerial tasks. We propose that tailoring this human-AI interaction to align with individuals' cognitive preferences is essential for enhancing performance. This hypothesis is examined through a field experiment in a multinational pharmaceutical firm. In the experiment, we manipulated four contextual parameters of human-AI interaction—work procedures, decision-making authority, training, and incentives—to align with sales experts' cognitive styles, categorized as either adaptors or innovators. Our results show that tailored interaction significantly improves sales performance, whereas untailored interaction results in negative treatment effects compared with both the tailored and control conditions. Qualitative evidence suggests that this negative outcome arises from role conflicts and ambiguities in untailored interaction. Exploring the mechanisms underlying these outcomes further, a mediation analysis of AI login data reveals that human-AI interaction tailoring leads sales experts to adjust their AI utilization, which contributes to the observed performance outcomes. These findings support a human-centered approach to AI that prioritizes individuals' information-processing needs and tailors their interaction with AI accordingly.

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Keywords: artificial intelligence • algorithms • behavioral theory • decision making • field experiment

1. Introduction

Recent management research has focused on using artificial intelligence (AI) for automation, with the ultimate goal of replacing humans entirely (Kesavan and Kushwaha 2020, Sun et al. 2022). However, for unstructured managerial tasks[1] involving uncertainty, automation is generally unfeasible (Raisch and Krakowski 2021, Boyacı et al. 2024). These tasks require interaction between humans and AI—known as augmentation—to leverage their complementary information-processing abilities, which is expected to enhance performance (Choudhury et al. 2020). Nonetheless, empirical evidence on the performance impact of augmentation remains inconclusive (Kim et al. 2024, Hillebrand et al. 2025).

We propose that tailoring human-AI interaction[2] to align with individuals' cognitive preferences is essential for enhancing performance in unstructured decision-making tasks. Microcontingency arguments suggest that humans require a supportive context aligned with their cognitive styles to process information effectively. To test this argument, we conducted a field experiment in a multinational pharmaceutical company that implemented an AI-based sales system. At the business-unit level, we manipulated four contextual parameters—work

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procedures, decision-making authority, training, and incentives—to influence how sales experts interacted with AI, while keeping the AI system itself constant across treatment conditions. We compared a tailored interaction group, where these four parameters aligned with sales experts' individual cognitive styles (Kirton 1976, Carnabuci and Diószegi 2015), to an untailored interaction group and a control group using the legacy IT system, a basic information repository.

Using a difference-in-differences (DiD) approach, we find that untailored interaction with AI negatively impacts performance compared with the legacy IT system, contrary to expectations. In contrast, tailored interaction yields positive effects relative to both the untailored and control conditions. Qualitative evidence indicates that while sales experts generally praised the system's quality and functionality, its introduction also caused major disruptions to experts' long-refined work procedures, especially since AI encroaches on many traditional human tasks. In the untailored condition, AI's benefits could not overcome these disruptions, which is also reflected in a gradual decline in AI system utilization in this condition. Given the relatively small number of clusters in our experiment, we conducted multiple sensitivity analyses, with consistent results supporting the robustness of our treatment effects. However, we caution against causal interpretations due to power constraints. Overall, our results highlight the importance of tailoring human-AI interaction to align with human needs in leveraging complementary information-processing capabilities. We contribute a *human-centered approach to AI* that contrasts with prior work emphasizing AI as a replacement for humans.

2. Theoretical Background and Hypotheses

Driven by the proliferation of data and advances in computational power, AI algorithms are increasingly being integrated into organizations (Benbya et al. 2020). AI encompasses technologies that enable machines to perform cognitive tasks traditionally associated with humans, such as reasoning, learning, and decision-making (Rai et al. 2019, Krakowski et al. 2023). Unlike earlier technologies, contemporary AI algorithms can learn autonomously and adapt their behavior through machine learning, leveraging data exposure without requiring explicit human programming (Mitchell 1997, Chalmers et al. 2021).

AI algorithms offer information-processing capabilities that complement those of humans. While AI excels in capacity, speed, and consistency in numeric computation, humans are better equipped for perceptual and associative tasks, such as recognizing emotions and providing contextual understanding (Agrawal et al. 2018, Choudhury et al. 2020). Additionally, humans can draw on valuable tacit or private information that remains inaccessible to AI (Ibrahim et al. 2021, Boyacı et al. 2024). Organizations have typically applied AI to structured tasks, aiming for automation and reduced human bias (Kesavan and Kushwaha 2020, Sun et al. 2022). However, for unstructured managerial tasks involving significant uncertainty, human-AI interaction is essential, as AI alone cannot deliver optimal solutions (Fortnow 2013). Scholars therefore advocate for an augmentation approach that integrates human and AI capabilities (Murray et al. 2021, Raisch and Krakowski 2021).

The augmentation literature suggests that human-AI interaction in unstructured decision-making tasks leads to higher performance (Choudhury et al. 2020, Lebovitz et al. 2022). AI's speed and accuracy help identify patterns that may elude human analysis, mitigating cognitive biases (Krakowski et al. 2023). Humans, in turn, contribute contextual understanding, intuition, and ethical judgment, which are essential in uncertain environments (Murray et al. 2021, Raisch and Fomina 2025). However, scholars acknowledge potential drawbacks of human-AI interaction. Performance benefits may be constrained by role ambiguities and conflicts (Chen et al. 2022, Dennis et al. 2023). Behavioral biases, such as algorithm aversion, may lead experts to ignore AI outputs (Dietvorst et al. 2018). Empirical research has yielded inconclusive results, with some studies reporting positive outcomes (Allen and Choudhury 2022, Chen et al. 2022), while others present negative or mixed findings (Fügener et al. 2022, Kim et al. 2024). Given these discrepancies, we reassess the anticipated positive relationship between human-AI interaction and decision performance.

Hypothesis 1. *The combination of human and AI information processing, all else equal, increases decision performance.*

We propose that the complementarities between human and AI information processing are dependent on tailoring the interaction context to align with decision-makers' cognitive needs. Drawing on organization design and behavioral theory, the microcontingency perspective emphasizes the importance of "complementary fit," or congruence between key contextual parameters (e.g., authority and incentives) and individuals' cognitive styles, which reflect persistent differences in how people organize and process information (Cable and Edwards 2004, Ostroff 2012). A prominent conceptualization of cognitive styles is Kirton's (1976) distinction between adaptors and innovators, based on the Kirton Adaption-Innovation Inventory (KAI).[3] While adaptors prioritize precision, structure, and methodical work, preferring to "do things better," innovators value variation, ambiguity, and flexibility, preferring to "do things differently" (Kirton 1976, p. 622; Carnabuci and Diószegi 2015, p. 883).

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Prior research suggests that tailoring work contexts to these cognitive styles improves decision-making outcomes. Four key parameters—work procedures (Kirton et al. 1991), decision-making authority (Hunt et al. 1989), training (Green and Hughes 1986), and incentives (Baer et al. 2003)—can be tailored to align with individuals' cognitive styles. For adaptors, interacting with an AI system is most effective when procedures are well-defined, decision-making authority is constrained, training is mandatory, and incentives are extrinsic, as these reduce ambiguity and offer clear guidance. Innovators, on the other hand, perform better with flexible procedures, high decision-making authority, on-demand training, and intrinsic incentives. In both cases, such tailoring can mitigate potential disruptions (Chen et al. 2022) and improve human information processing (Kirton et al. 1991), thereby enhancing decision-making performance.

Hypothesis 2. *Tailoring the human-AI interaction context (i.e., procedure, authority, training, incentives) to humans' cognitive styles increases decision performance.*

3. Empirical Context and Human-AI Interaction

We conducted our empirical study in the four Nordic sales subsidiaries (i.e., Denmark, Finland, Norway, and Sweden) of a leading global pharmaceutical company. Due to restrictive regulations, pharmaceutical firms in this region primarily compete through indirect drug sales via prescribers, typically medical doctors, rather than by directly raising patient awareness through promotion or managing healthcare centers. Selling drugs in this context constitutes a highly unstructured task environment, given the complexity of the product portfolio and the diverse personalities, motivations, and preferences of prescribers.

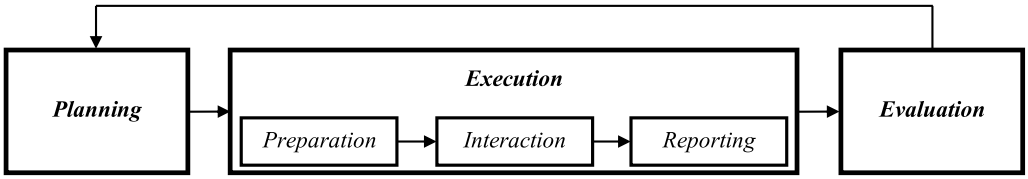
In sales meetings, sales experts provide prescribers with information about the company's drugs, including medical efficacy profiles, side effects, interaction effects, and comparisons to rival products. Conducting these meetings is essential for future sales, making these meetings a reliable proxy for anticipated sales (Brown and Peterson 1994, Ahearne et al. 2010, Chung et al. 2019). Sales meetings are highly idiosyncratic, with outcomes heavily dependent on the sales experts' decision making, such as prioritizing clients, setting sales goals, allocating time and resources, and tailoring information and messaging. Traditionally, sales experts prepared and conducted these meetings using their expertise and client and product information stored in a basic repository.

In 2014, the Nordic subsidiaries introduced a state-of-the-art AI sales system to guide sales experts. At the time, the sales system was one of the first AI algorithms globally to be applied to unstructured decision-making tasks in organizations. The AI algorithm's application in sales spans three stages of the sales process: *planning*, *execution*, and *evaluation* (see Figure 1). During planning, the system generates lists of clients to target and suggests sales objectives. Execution involves preparing for and conducting meetings while reporting outcomes back to the system. Evaluation compares planned versus achieved goals and informs future planning.

Due to the unstructured nature of pharmaceutical sales, achieving competitive results requires substantial tacit knowledge, creativity, and judgment. For example, a top-performing sales expert in our sample relied extensively on memory, judgment, and improvisation during meetings, using personal details about clients, attuning to their emotional cues, and engaging in informal dialogue to enhance performance. Such tacit, rare, and idiosyncratic information is challenging to codify, making it inaccessible to algorithms and reliant on human perceptual and subconscious judgment (Brynjolfsson and Mitchell 2017, Boyacı et al. 2024, Kim et al. 2024). At the same time, cross-client patterns can significantly influence sales performance. In our setting, an average sales expert interacts with over 100 client organizations, each with hundreds or even thousands of prescribers. Human cognitive limitations make it impossible to comprehensively analyze such vast and complex company-wide data. However, AI algorithms excel at processing this type of information quickly, consistently, and comprehensively. Such AI processing can provide sales experts with previously unavailable codified insights, including ranked lists of clients to target for maximizing sales outcomes and tailored sales messages to match clients' individual preferences.

One of the authors spent several days in the field observing sales experts' interactions with AI. These observations, corroborated by subsequent discussions

Figure 1. Sales Process and Substages



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with senior management, revealed that sales experts adopted the AI system's recommendations, such as client prioritization and sales messages, in the vast majority of cases. Although they could have relied minimally on the AI system following the experimental intervention—particularly in one condition (allocation to Design 3, D3), where innovators were only required to record basic meeting details such as date and outcome—this option was rarely exercised. Sales experts widely praised the AI system's capabilities and deviated from its suggestions only in exceptional cases, typically when they had private information, needed to complement the AI system's outputs with tacit insights, or were faced with circumstances that required adjustments. For example, sales experts sometimes altered client targeting when they unexpectedly gained access to prescribers or received ad hoc personal referrals.

4. Research Design, Data, and Method

The pharmaceutical company introduced the same AI sales system across its Nordic regional markets but delegated its implementation to each country subsidiary, enabling this field experiment. The experiment covered sales activities for four specialty drugs launched in the Nordic markets between 2007 and 2011: Product A (rheumatic), Product B (neurological), Product C (pulmonary), and Product D (gastrointestinal). Dedicated business units in Denmark, Finland, Norway, and Sweden managed each product, supported by substantial sales forces in each market.

The field experiment included 72 sales experts across 12 business units, which served as the unit of randomization, and ran from Q1/2012 to Q2/2017. During this time, the company tracked meetings and sales data quarterly. The initial period from Q1/2012 to Q1/2014 (9 quarters, or 2 years and 3 months) served as a baseline monitoring phase without intervention. The AI system was implemented in Q2/2014, and monitoring continued until Q2/2017 (13 quarters, or 3 years and 3 months), at which point the experiment concluded. In total, the observation period spanned 22 quarters, or 5.5 years.

We implemented the experiment at the business-unit level, as this approach best balanced the dual objectives of minimizing spillover risk and enabling stratification to account for country- and product-level differences. The latter was crucial because the pharmaceutical industry is highly regulated, with many idiosyncrasies at both the country and product levels. To account for these idiosyncrasies, business units are organized in a highly decentralized manner, with almost no collaboration across business units. Although spillovers cannot be categorically ruled out, sales experts in the Nordic pharmaceuticals industry describe themselves as "lonely professionals" who spend most of their time traveling alone in their cars and visiting various, often remote, client locations. They do not work in teams and typically visit the company's offices no more than one day per month. While biannual sales conferences bring together sales experts across business units, these conferences have a busy schedule with intensive workshops focused on specific products and sales training. Collectively, these factors significantly reduce the risk of spillovers when conducting the experiment at the business-unit level.

We used stratified random sampling to assign the 12 business units to three groups, each adopting one of three experimental designs (see Tables 1 and 2). The strata included the aforementioned products and countries. The designs were allocated as follows: Design 1 was implemented in Denmark (Product A), Finland (Product B), Norway (Product C), and Sweden (Product D). Design 2 was implemented in Denmark (Product D), Finland (Product A), Norway (Product B), and Sweden (Product C). Finally, Design 3 was implemented in Denmark (Product B), Finland (Product C), Norway (Product D), and Sweden (Product A). The remaining four business units were used by the company to test an alternative condition not included in this study.

While the company informed sales experts about the introduction of the new sales system and its evaluation, they were not made aware of the particularities of the study design, such as their allocation to control and treatment groups. Online Appendix A provides a detailed description of the data collection, experiment instructions, and subject eligibility, while Online Appendix B contains a graphic evaluation and analysis of sample size and statistical power (indicating the study is underpowered, with 37% power for D2 and 61% for D3). With all participants in the experiment successfully registered across all time periods,[4] the data set constitutes a balanced panel comprising 1,584 observations.

4.1. Dependent Variables

4.1.1. Sales Performance. Our primary dependent variable is the number of *sales meetings*, a key performance metric for pharmaceutical companies in the Nordic region. Due to strict regulations, prescriber access is highly constrained, making the ability to schedule and conduct meetings a strong predictor of future sales (Brown and Peterson 1994, Dylst et al. 2011, Chung et al. 2019). Through improved information processing, sales experts can better anticipate and address prescribers' needs and preferences, thereby directly enhancing subsequent sales performance. The AI system was introduced to support this capability by facilitating the arrangement and execution of

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Table 1. Detailed Parameter Descriptions

Work procedure	Imposed	Sales experts have to follow a comprehensive, predefined process for interacting with the AI system. This process includes the generation of a client target list, the development of a "customer interaction plan," and the generation of summary client reports. Compliance with the interaction process is monitored by a manager.
	Flexible	Sales experts can adjust the above-described interaction process freely (without monitoring by a manager). The only mandatory element of interaction pertains to the submission of a final sales meeting summary report (which allows for machine learning).
Decision-making authority	Low	Sales experts have to adhere to the AI system's budgetary constraints (with any deviation requiring managerial approval).
	High	Sales experts can, within certain limits, deviate from the AI system's resource allocation suggestions (without managerial approval).
Training	Push	Sales experts take a standardized, in-person training course that explains the AI system's functionality and its use in the sales process. The training course is conducted through an introductory session prior to system introduction and a follow-up session two months later.
	Pull	Sales experts can customize the training in terms of format, content, and timing according to their individual preferences. On-demand training is offered via online tools and phone support.
Incentives	Extrinsic	Sales experts are extrinsically incentivized with an annual monetary bonus based on their use of the system and their subsequent sales performance.[a]
	Intrinsic	Sales experts are provided with intrinsic motivation in terms of social appreciation and attention by their managers.[b] This treatment included up to three times more feedback meetings with their superior, including cotraveling to client sites.

[a]Before the experimental intervention, the bonus could amount to the equivalent of four months' salary. Following the intervention, the amount could increase to the equivalent of six months' salary.
[b]Sales experts kept their monetary incentive of four months' salary.

successful meetings. We measured this variable as the average daily number of client meetings per quarter.[5]

As a secondary performance measure, we used the number of *units sold*, a reliable proxy for sales performance due to minimal price competition and standardized pricing in the Nordic region.[6] While this measure more accurately reflects overall market performance, it is less directly tied to sales experts' decisions. Public-sector procurement decisions, often bundled and influenced by long-term regulatory constraints, result in delayed and indirect performance outcomes. To account for these delays, we conducted a supplementary analysis (see Online Appendix D) by lagging the *units sold* variable by 2, 4, 6, and 8 quarters (ranging from half a year to two years). This variable was measured as the average daily number of product units sold within a given quarter.

4.2. Main Independent Variables

4.2.1. Treatment Design. We used a categorical variable to represent the three treatment designs, which we interacted with a binary pre/post variable to estimate the treatment effect (see the "Model specification" section for further details). As shown in Table 2, Design 1 (D1) serves as the control group, Design 2 (D2) represents an untailored interaction with the AI system,[7] and Design 3 (D3) captures the tailored interaction, where the four parameters (i.e., work procedure,

Table 2. Allocation of Experimental Designs

	Design 1 (D1)	Design 2 (D2)	Design 3 (D3)	
	Control group	Untailored interaction	Tailored interaction[a]	
Sales experts	All	All	Adaptors	Innovators
Sales system	Legacy	AI	AI	
Interaction-context tailoring	No	No	Yes	
Work procedure	n/a	Flexible	Imposed	Flexible
Decision-making authority	n/a	Low	Low	High
Training	n/a	Push	Push	Pull
Incentives	n/a	Extrinsic	Extrinsic	Intrinsic

[a]The system tailoring was conducted based on focal sales expert cognitive style to provide a theoretically optimal fit. For this reason, a distinction is made with regard to adaptors (left-hand side column) and innovators (right-hand side column) within the same experimental treatment (i.e., D3).

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decision-making authority, training, and incentives) were aligned with each sales expert's cognitive style (i.e., adaptor or innovator).

4.2.2. Cognitive Style. We used an established 32-item questionnaire (Kirton 1976) with 5-point Likert-style response options ranging from "strongly agree" to "strongly disagree" (Likert 1932) to measure the sales experts' *cognitive styles*. The results, collected for all sales experts prior to the experimental intervention, were used to calculate the KAI index, which positions respondents along the "adaptive-innovative" continuum. By summarizing the 32 responses following established methodology (Kirton 1976), we obtained a sample of individuals who ranged from 72 to 119 (with a theoretical population mean of 96 and a sample mean of 91.92). Consistent with the literature (Foxall 1995, Amabile et al. 2001), we classified sales experts with a KAI score below 96 as "adaptors" (47 individuals) and those with a score above 96 as "innovators" (25 individuals). To ensure the robustness of our findings and further explore heterogeneity across the KAI spectrum, we conducted a supplementary analysis where we segmented the sales experts into four KAI-based categories. The results are available in Online Appendix E.

5. Results

5.1. Descriptive Statistics and Preliminary Checks

Table F1 in Online Appendix F provides summary statistics for the 72 sales experts across the 12 business units, showing a balanced distribution of gender, age, education, and work experience. Table 3 provides summary statistics categorized by experimental design. While demographic characteristics did not differ significantly across designs, baseline performance measures exhibited some variation. However, F-tests confirmed that these variables were not jointly significant in distinguishing between the designs.

Table 4 presents the descriptive patterns for *sales meetings* and *units sold* before and after the intervention. In the control group (D1), *sales meetings* decreased slightly postintervention. Sales experts in D2 *reduced* their daily *sales meetings* by 0.46 compared with the control group, while those in D3 *increased* their daily meetings by 0.84. Similar patterns were observed for *units sold*.

Figure F1, (a) and (b) in Online Appendix F depict performance trajectories, showing the control group (D1) remaining stable over time, D2 declining after the Q2/2014 intervention, and D3 improving. Table 4 indicates pretreatment differences, with *sales meetings*

Table 3. Descriptive Statistics of the Sales Experts and Balance Checks (Q1/2012)

	Mean (SD)			*t* statistics (SD)	
Variable	D1	D2	D3	(D2–D1)	(D3–D1)
Gender	0.57	0.58	0.52	0.08	−0.36
	(0.51)	(0.50)	(0.51)	(1.01)	(1.02)
Age	34.71	35.29	34.19	0.32	−0.32
	(6.08)	(5.89)	(5.52)	(11.99)	(11.63)
Marital status	0.67	0.63	0.74	−0.29	0.55
	(0.48)	(0.49)	(0.45)	(0.98)	(0.93)
Education	1.67	1.71	1.52	0.22	−0.88
	(0.66)	(0.62)	(0.51)	(1.28)	(1.17)
KAI cognitive style	93.14	93.13	89.89	−0.01	−1.10
	(11.26)	(11.13)	(9.31)	(22.44)	(20.57)
Years of experience in industry	10.81	10.29	9.04	−0.31	−1.11
	(6.27)	(4.74)	(4.77)	(11.03)	(11.03)
Years of experience in sales	6.95	6.38	6.74	−0.56	−0.18
	(4.42)	(2.37)	(3.65)	(6.97)	(8.07)
Years of experience in the company	4.48	4.71	4.67	0.44	0.32
	(2.27)	(1.20)	(1.84)	(3.57)	(4.11)
Baseline sales meetings	2.14	2.05	1.63	−0.57	−3.01***
(Q1/2012–Q1/2014)	(0.58)	(0.56)	(0.59)	(1.15)	(1.18)
Baseline units sold	1,784.18	3,862.31	1,669.86	2.36**	−0.27
(Q1/2012–Q1/2014)	(1,604.89)	(3,750.06)	(1,318.79)	(5,919.11)	(2,923.20)
F-test of joint significance (F-stat)				0.78	1.60
N (KAI Adaptors)	11	16	20		
N (KAI Innovators)	10	8	7		
N (Total)	21	24	27	45	48

Notes. Gender: 0 = male, 1 = female. Marital status: 0 = single, 1 = married. Education: 1 = bachelor's degree, 2 = master's degree, 3 = doctoral degree.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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Table 4. Pre- vs. Posttreatment Performance, DiD (Descriptive)

Outcome variable	Before	After	Difference	DiD
Meetings				
D1/Control group	2.14	1.95	−0.20	
Adaptors	2.54	2.55	0.01	
Innovators	1.71	1.29	−0.42	
D2	2.05	1.39	−0.65	−0.46
Adaptors	2.34	1.52	−0.82	−0.83
Innovators	1.45	1.15	−0.31	0.12
D3	1.63	2.28	0.65	0.84
Adaptors	1.69	2.46	0.78	0.77
Innovators	1.47	1.75	0.28	0.70
Sales				
D1/Control group	1,784.18	2,018.14	233.96	
Adaptors	2,619.66	3,030.82	411.16	
Innovators	865.16	904.20	39.04	
D2	3,862.31	3,548.95	−313.36	−547.32
Adaptors	4,722.78	4,271.09	−451.69	−862.85
Innovators	2,141.38	2,104.67	−36.70	−75.75
D3	1,669.86	2,353.34	683.48	449.52
Adaptors	1,955.35	2,787.21	831.86	420.70
Innovators	854.17	1,113.71	259.54	220.50

lower in D3 and *units sold* higher in D2 compared with other groups. Our DiD analysis does not require identical pretreatment levels across groups but relies on the common-trends assumption, which posits similar trends in the absence of intervention (Angrist and Pischke 2009, p. 171f; Wing et al. 2018). Figure F1, (a) and (b) confirm stable pretreatment trends across groups, substantiating random allocation of business units and satisfying the assumption. A supplementary robustness test related to the common-trends assumption is available in Online Appendix G.

Figure F2 in Online Appendix F shows no notable market fluctuations that could bias the results. Consistent with these trends, Figure F3 shows a stable market share for the control group (D1), a decline for D2, and a gradual increase for D3 postintervention. The stability of these metrics over the pretreatment period (9 quarters) further reinforces the credibility of our experimental design. Separate analyses by country and product yield consistent results across subgroups (Online Appendix H), confirming that sales experts in D3 outperformed those in D2 after the intervention, regardless of market or product.

Descriptive evidence suggests that innovators generally underperformed compared with adaptors across all designs (Table 4). Innovators are underrepresented in the sample, with an average KAI score of 92, compared with the theoretical mean of 96. This aligns with the highly technical and regulated nature of the pharmaceutical industry, which prioritizes precision and consistency over creativity—traits more closely associated with adaptors. This self-selection mechanism likely explains both the overrepresentation of adaptors and their comparatively higher performance in this context.

To assess potential multicollinearity (Gujarati 1995), we analyzed pairwise correlation coefficients (Table F2, Online Appendix F). High correlations were limited to related variables, such as educational attainment and work experience. Only *age* and *industry experience* exceeded the conventional correlation threshold of 0.80 (Berry and Feldman 1985). Variance inflation factors (VIF) and tolerance diagnostics (Table F3, Online Appendix F) revealed no serious multicollinearity issues when omitting *age*, with a mean VIF of 2.41 and a maximum of 6.71, both below critical thresholds (Neter et al. 1990, Hair et al. 1998). Due to their high correlation, *age* was excluded from regression models, while *industry experience* was retained to account for linear trends through fixed effects.[8]

5.2. Model Specification

Given the experimental setup and resulting panel data structure, we employed panel data methods using ordinary least squares (OLS) estimation (Angrist and Pischke 2009, p. 165ff). To assess the effects of the different designs on sales performance, we used a DiD linear model (Angrist and Pischke 2009, p. 169ff) controlling for time-fixed effects (i.e., current year and quarter) and clustering standard errors at the business-unit level, the unit of randomization in the experiment (Abadie et al. 2023). Using *sales meetings* and *units sold* as the dependent variables, we examined the effect of the different designs (with the control group D1 as the baseline) by estimating the multiplicative interaction terms formed by the dummy variables for treatment design and an indicator variable for pre- or posttreatment. The three interaction variables are our primary independent variables of interest. Further, we

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controlled for additional observable variables, namely work *experience* (years worked in the pharmaceutical *industry*, in pharmaceutical *sales*, and at the focal *company*), *gender*, *marital status*, *education*, *KAI cognitive style*, and *product handled*. The estimation followed this form:

$$Y_{igt} = \alpha_i + \gamma_t + \lambda_q + \beta_1 \cdot post_t + \beta_2 \cdot D2_g + \beta_3 \cdot D3_g + \beta_4 \cdot (D2_g \times post_t) + \beta_5 \cdot (D3_g \times post_t) + X'_{igt} \cdot \delta + \varepsilon_{igt}$$

where:

Y_{igt} denotes the performance outcome for individual i in group g at time t

α_i represents individual fixed effects

γ_t represents year fixed effects

λ_q represents quarter fixed effects

$post_t$ is an indicator for the posttreatment period (with coefficient β_1 representing the general time trend effect for the posttreatment period)

$D2_g$ and $D3_g$ are indicators for groups D2 and D3, respectively (with respective coefficients β_2 and β_3 designating the time-invariant differences between groups D2 and D3 compared with the control group D1)

β_4 and β_5 are the DiD estimators that designate the difference in changes over time (i.e., the coefficients of interest) for groups D2 and D3, respectively

X'_{igt} is a vector of additional time-varying control variables (with coefficients δ representing the effects of the control variables on the outcome)

ε_{igt} denotes the random error term, clustered at the business-unit level (assumed to be independently and identically distributed, aside from the clustering at the business-unit level)

We controlled for the aforementioned covariates using fixed effects, and where applicable, time-invariant covariates such as *design*, *gender*, *and KAI cognitive style* were dropped from the estimations. To evaluate the choice between fixed-effects and random-effects models, we conducted a Sargan-Hansen test (Sargan 1958, Hansen 1982). Unlike the Hausman (1978) test, the Sargan-Hansen test accommodates heteroskedastic- and cluster-robust standard errors when comparing model specifications (Arellano 1993). The test was implemented in Stata using Baum et al.'s (1999) "xtoverid" command. Testing the orthogonality assumption of the random-effects model significantly ($p < 0.001$) rejects the null hypothesis that regressors are individual-level effects uncorrelated with the independent variables. This result indicated that fixed-effects estimation is preferable for both dependent variables. The results from the Sargan-Hansen test of overidentifying restrictions for the two outcome variables are provided in Table F4 in Online Appendix F.

5.3. Main Analysis

Table 5 presents the main treatment effects and the results of a sensitivity analysis, including cluster-robust inference and randomization inference (see Online Appendix J for a detailed discussion). On average, sales experts in D2 conducted 0.46 *fewer* daily meetings than those in the control group. In contrast, sales experts in D3 conducted 0.84 *additional* daily

Table 5. Main Treatment Effects

Dependent variable	Sales meetings	Units sold
D2 · Post	−0.46***	−547.32*
	(0.06)	(186.74)
	{<0.001}	{0.012}
	[0.010]	[0.006]
	⟨0.0260⟩	⟨0.0260⟩
D3 · Post	0.84***	449.52
	(0.05)	(323.27)
	{<0.001}	{0.232}
	[<0.001]	[0.266]
	⟨0.0270⟩	⟨0.2680⟩
Post	−0.18	142.46
	(0.18)	(251.36)
Experience	−0.02	32.31
	(0.06)	(89.17)
Constant	2.14**	1,975.19
	(0.63)	(943.23)
Fixed effects (coefficients omitted for brevity)	*Individual, Year, Quarter*	*Individual, Year, Quarter*
N	1,584	1,584

Notes. Fixed-effects panel estimation, standard errors clustered at the business-unit level. Robust standard errors in round brackets. *p*-values obtained from WCR bootstrap clustering with 1,000 replications in curly brackets. *p*-values obtained from jackknifed wild bootstrap clustering with 1,000 replications in square brackets. RI-based *p*-value from 1,000 resamplings in angle brackets.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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meetings compared with the control group. For the average sales expert, this represents a decrease of approximately 20% in D2, while D3 shows an increase of more than 40%.

Estimating the same models with *units sold* produces similar results in both direction and magnitude. Sales experts in D2 performed worse following the AI system's implementation, while those in D3 outperformed the control group. On average, sales experts in D2 sold approximately 550 *fewer* units per day, whereas those in D3 sold nearly 450 *additional* units daily. This indicates that for the average sales expert, D2 led to a decline of around 20% in sales performance, while D3 yielded an increase of 16%, showing that the increase in *sales meetings* translates into additional *units sold*. We also conducted ANOVA pairwise comparisons (results available in Online Appendix K) of mean performance between designs, before and after treatment, respectively (see Tables K1 and K2), as well as pairwise direct comparisons of the obtained coefficients within each group (pre- versus posttreatment) and against each other (see Table K3). Results confirmed that D2 is associated with a significant decrease, and D3 with a significant increase, in *sales meetings*. For *units sold*, the change is statistically significant for D2 but not for D3.

Given that we have a linear specification in our regression model, we tested for time-dependent effects after the treatment intervention using point estimates for expected quarterly performance. The plotted estimates, available in Online Appendix L (Figure L1 for *sales meetings* and Figure L2 for *units sold*), revealed no significant time-varying effects influencing our linear models. This indicates no evidence of habituation effects following the implementation of the AI system.

We conducted supplementary analysis introducing time lags in the *units sold* variable to account for the expected delayed and intermittent manifestation of the intervention effects. The results (available in Online Appendix D) show that the effects for D2 remained significant up to 1.5 years posttreatment, increasing in magnitude over time to 1.84 times the immediate effect. For D3, the effects became significant after 1 year and remained significant for up to 2 years, also increasing in magnitude over time (reaching 1.48 times the immediate effect). These findings suggest that the treatment effects unfold over time, with negative effects materializing more immediately and positive effects appearing after a delay.

To further explore the robustness of the main findings and to rule out possible inference issues (Athey and Imbens 2017), we performed our analysis at the cluster level in addition to the individual level. The results are available in Online Appendix M and show general consistency, with the notable exception that there are not only significant results for *sales meetings* (our primary dependent variable) but also for *units sold* (our secondary dependent variable) in D2 and D3. The magnitude of the estimate for D3 becomes substantially larger, showing a significant increase of approximately 715 daily units sold per sales expert, compared with around 450 (and nonsignificant) in the individual-level analysis. Furthermore, we used an ANCOVA specification in line with McKenzie (2012), employing baseline variables for *sales meetings* and *units sold* that capture the mean of the pretreatment observations for each salesperson. The results (see Table N1 in Online Appendix N) are highly consistent with the DiD analysis in terms of magnitude and significance. In addition, as our outcome variables are based on the averaged quarterly count of *sales meetings* and *units sold*, we conducted a robustness check using alternative Poisson model specifications. These findings (see Table O1 in Online Appendix O) are internally consistent and aligned with those from our main DiD specification in terms of direction and significance levels.

5.4. Counterintuitive Findings and Mechanisms

Parts of the quantitative results align with our theory and hypothesized outcomes, while the negative treatment effect of D2 does not. To explore this discrepancy and gain deeper insight into the mechanisms driving the observed treatment effects, we draw on complementary qualitative data from postexperimental interviews and quantitative AI-system login data. Additionally, we consider alternative explanations, such as the possibility that new work practices introduced alongside the AI system—rather than their influence on AI system utilization—may account for the observed treatment effects. We believe these supplementary analyses and explanations provide further support for our main theory on AI interaction tailoring.

Concerning the substantial treatment effects, sales experts reported in postexperiment interviews that the introduction of the AI system had a profound impact on their work, both through reconfigured work procedures and the utility of the AI system itself. For instance, a Danish innovator remarked: "We received the best work tools I could imagine! The difference compared with the old IT system and way of working before the change was radical for me!" Many sales experts highlighted information-processing complementarities, a central thesis of our study and a key motivation behind the adoption of human-AI interaction (Boyacı et al. 2024). Large-scale analytical tasks were now handled by the AI system, which provided the sales experts with more information than before and allowed them to redirect their freed-up time. Accordingly, a Norwegian innovator explained: "The new way of working, thanks to the IT system, gave

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me more time to handle the challenge of how to access the client." Ideally, this human-AI interaction created a more effective division of labor. For example, a Finnish adaptor reflected: "If I compare a day of work in the old way and a day of work in the new way, I think I did more than double as much relevant work after the change."

Consistent with these examples, sales experts generally praised the AI system's quality, functionality, and complementarities with their own skills and information. However, those in the untailored design (D2) also reported negative experiences, including repeated disruptions, ambiguities, and role conflicts stemming from the AI system's implementation. Sales experts attributed these challenges to the significant adjustments required to their daily work. For instance, a Danish innovator in D2 acknowledged the AI system's sophistication but also noted: "The whole work situation was in a way paradoxical. We got this super tool, and at the same time, I felt like in prison. There was no freedom to work the way I wanted to work. (…) I wanted to work more spontaneously and in an improvised way to deal with each situation. This level of controlling killed my internal drive!" Similarly, a Norwegian innovator in D2 observed: "The IT system was great, it had much more valuable information. However, I wanted to use it in my own way. The whole change made me lose my belief that I could reach the high sales objectives set for me. In a way, after the change, I felt like a less good sales representative, because of that prison-like context."

A key insight from our quantitative analysis is that much of the negative treatment effect in D2 originates from adaptors (see Table 4). This is surprising, as only one parameter—work procedure—was misaligned with adaptors' needs in the untailored implementation, while the remaining three parameters were aligned (making the design appear less suitable for innovators). However, based on our observations and interview data, work procedure emerged as the most critical parameter, shaping daily operational interactions between the sales experts and the AI system.[9] For adaptors, an imposed work procedure would have been especially valuable, providing the necessary guidance and structure in a work environment marked by significant ambiguity. A Swedish adaptor assigned in D2 expressed the need for more structured work procedures, stating: "With that new IT system we should have developed a new way of doing our work. Without such guidance, the new IT system became to me more of a distraction that took time and effort (…) that was cool (…) yet never realized (…) benefit for my work." Other adaptors shared similar feedback, often describing the AI system as "great" but noting that the flexible work procedures left them "unable to make good use of [the AI system]" (Finnish adaptor). A Swedish adaptor in D2 reflected: "Nearly all was good with that change except that I felt that it was messy to work with (…) then I did not put much effort into my work and into mastering the new IT system and its functionalities."

Conversely, adaptors in D3 praised the clear guidelines and structured procedures for working with the AI system. A Swedish adaptor in D3 noted: "Everything with the new way of working was very clear, well organized, and well structured, which was great to me." Similarly, a Norwegian adaptor highlighted: "We received excellent guidelines for how to work and how to use the new IT system. This made me feel good; I could focus better on the content of my work, namely the customers. Somehow, I became better at doing my work, as the system and the sales process guided me well, which felt good. It made me relax. I knew what to do and how to do it." Innovators in D3 also praised the system's functionality and the tailoring of system interactions to their preferences. For instance, a Swedish innovator in D3 remarked: "We received excellent training, which was great as I could train when I needed it and specifically focus on what I needed. I became more flexible in my work and could work more in line with my intuition rather than following strict procedures."

One challenge of our experimental design lies in isolating our main theoretical argument—improved human-AI interaction through parameter tailoring—from its underlying components, namely the effects of modified work parameters versus the AI system itself. This difficulty arises because the work parameters were adjusted alongside the introduction of the AI system, making it hard to isolate the precise source of the observed treatment effects. It could be argued that the four adjusted parameters—particularly work procedure—are the primary drivers of our results, rather than the AI system and its utilization. However, while fully separating these effects would be theoretically desirable, doing so is practically challenging given AI's tendency to encroach on areas of human expertise, necessarily entailing changes in contextual factors such as work procedures and decision-making authority. We see this as a distinguishing characteristic of AI systems, which—unlike previous IT solutions that typically operated as isolated systems—require defining implementation parameters that reshape the broader work context (Ångström et al. 2023, Kemp 2024). Accordingly, this study focuses on AI *implementation* rather than AI *in isolation*, recognizing that it is practically impossible to separate AI from the contextual parameters shaping its use. This broader impact likely explains the substantial treatment effects observed in our study.

To address the question of whether our results are primarily driven by changes to work parameters or their influence on AI utilization, we analyzed system login data, beginning with descriptive evidence and

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followed by a mediation analysis. This mediation analysis provided tentative evidence that the direct effects of our treatments on sales performance are mediated by AI system utilization when comparing D2 and D3.

While no login data were available for the legacy system (i.e., the control group), we descriptively compared the AI system login frequencies between D2 and D3 (average daily logins are provided in Table 6). For innovators, initial utilization numbers were high (Q3/2014), then declined (Q4/2014), and ultimately diverged between D2 and D3. While utilization gradually increased in D3, it continued to decrease in D2. This trend mirrors the performance results for D2 versus D3 in terms of both *sales meetings* and *units sold* (see Table F5 in the Online Appendix). This pattern is consistent for both the overall D2 and D3 treatment groups and the respective subgroups of adaptors and innovators within each group.[10] Although purely correlational, this analysis suggests that AI system utilization—rather than the work parameters, which remained constant throughout the postintervention period—is the primary driver of our results.

A more detailed analysis of system utilization shows an initial spike in innovators' login numbers. Taken together with our interview data, this reveals that innovators exhibited particularly high interest in the AI system early on. Subsequently, their system use depended on how well the interaction with the system suited their cognitive style, thus explaining the divergence in login frequencies between D2 and D3. A similar pattern emerged for adaptors, who also diverged in their system utilization between D2 and D3, though without an initial spike in curiosity. These additional findings suggest that the theoretically optimal tailoring of the interaction to align with sales experts' cognitive styles promotes greater AI utilization, which positively affects subsequent performance. Without tailoring, the human-AI interaction seems to impair sales experts' information processing, reducing their willingness to use the system or constraining their ability to benefit from it, ultimately leading to inferior sales performance.

To further explore the login dynamics and underlying mechanisms in D3, we conducted a multilevel mediation analysis based on Baron and Kenny's (1986) approach, testing the effect of D3 on *sales meetings* via logins (i.e., the mediator). This analysis focused on the posttreatment period for D3 (Q3/2014–Q2/2016), during which login data were available. We estimated the effects of D3 on *sales meetings* (Step 1), D3 on logins (Step 2), logins on *sales meetings* (Step 3), and finally D3 on *sales meetings* while controlling for logins (Step 4). We used a model with random intercepts across *country*, *business unit*, and *individual* levels, reflecting the nested data structure. The results presented in Table 7, which are based on 1,000 bootstrap replications for each estimation step, show that D3 significantly increases both *sales meetings* (1.05) and logins (2.03) relative to D2. Logins, in turn, positively impact *sales meetings* (0.41). In a comprehensive model including both D3 and logins as predictors, we find a significant direct effect. Finally, we computed the indirect and total effects, as well as the respective standard errors, using Stata's "nlcom" (nonlinear combination) command. The results are presented in Table 8 alongside those for the direct effect from Step 4. The findings from the mediation analysis imply that D3's influence on *sales meetings* is partially mediated through increased logins.

Aside from clarifying our mechanisms, the mediation analysis yields two additional insights. First, it corroborates the logic of our main analysis by documenting indirect effects. Specifically, the increased utilization of the AI system in D3—assuming its theoretical utility and gradual, autonomous improvement through machine learning enabled by more frequent logins—leads to improved sales performance compared with D2. However, it is important to acknowledge that the mediation analysis does not rule out the possibility of reverse causality; that is, the absence of tailoring may result in declining sales performance, which could subsequently reduce system use. A more detailed discussion of performance trends, changes in work conditions, and corresponding analyses of system usage over time is available in Online Appendix P.

Second, the significant residual direct effect of D3 on sales performance suggests the presence of a learning effect among individuals in D3. As shown in Table 7, D3 exhibits a step function improvement over D2, reflecting the primary AI-tailoring treatment. However, when examining the residual direct effect of D3 and its improvement over time in connection to cumulative logins (as depicted in Figure P1 in

Table 6. Average Daily Logins into the AI System

		Q3/2014	Q4/2014	Q1/2015	Q2/2015	Q3/2015	Q4/2015	Q1/2016	Q2/2016	Mean
D2	All	4.9	3.6	3.1	2.9	2.7	2.6	2.6	2.5	3.1
	Adaptors	5.3	4.1	3.4	3.1	2.8	2.8	2.8	2.8	3.4
	Innovators	4.1	2.7	2.6	2.5	2.4	2.3	2.3	2.1	2.6
D3	All	4.4	3.9	4.3	4.6	4.7	5.0	4.9	5.2	4.6
	Adaptors	4.4	4.5	4.8	5.2	5.1	5.4	5.4	5.7	5.1
	Innovators	4.4	2.4	2.9	3.1	3.3	3.7	3.6	3.7	3.4

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Table 7. Mediation Analysis (Mixed Model/Hierarchical Linear Modeling)

Dependent variable	Sales meetings Step 1	Logins Step 2	Sales meetings Step 3	Sales meetings Step 4
D3	1.05***	2.03***	n/a	0.22***
	(0.04)	(0.09)	n/a	(0.05)
Logins	n/a	n/a	0.41***	0.41***
	n/a	n/a	(0.02)	(0.02)
Experience	0.03***	0.06***	0.00	0.00
	(0.00)	(0.01)	(0.00)	(0.00)
Constant	0.91***	2.62***	−0.04	−0.15**
	(0.08)	(0.18)	(0.07)	(0.06)
Fixed effects (coefficients omitted for brevity)	*Year, Quarter*	*Year, Quarter*	*Year, Quarter*	*Year, Quarter*
Random effects (nested, coefficients omitted for brevity)	*Country, Business unit, Individual*	*Country, Business unit, Individual*	*Country, Business unit, Individual*	*Country, Business unit, Individual*
N	408	408	408	408

Note. Nested, robust standard errors obtained from bootstrap clustering with 1,000 replications in parentheses.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Online Appendix P), it becomes evident that sales experts in D3 gradually increased their system usage and improved their performance, indicating learning in their interaction with the AI system. This finding is particularly notable because the AI system itself is identical across D2 and D3, including its improvement over time via machine learning. Thus, the direct effect can be attributed to differences in sales expert behavior between D2 and D3 over time. In contrast, individuals in D2 gradually decreased their use of the AI system, likely as they recognized the limitations of their interaction with it. Overall, this supplementary analysis reveals a tentative learning effect among sales experts in D3 relative to D2, occurring alongside the previously established treatment effect and the gradual improvement of the AI system over time.

6. Discussion

Our field experiment revealed substantial positive (and negative) treatment effects associated with tailored (and untailored) human-AI interaction on unstructured decision-making tasks. These effects stem from the AI system's significant impact on sales experts, who concluded that the new system "changed everything" (Norwegian innovator) and became "central to their work" (Danish adaptor). Engaging with the AI system transformed those in the tailored interaction treatment group into "sales engines" (Swedish adaptor), while it "demotivated" and "frustrated" (Finnish adaptor) those in the untailored condition. Among the four treatment parameters, work procedure emerged as the most impactful, particularly for adaptors. Beyond the main treatment effects, our results suggest learning effects, showing no evidence of algorithm aversion in the tailored human-AI interaction. Moreover, we find that the negative consequences of untailored interaction manifest more quickly than the benefits of tailored interaction. Our findings remain robust across a range of robustness tests, though we caution against causal interpretation due to power constraints. Below, we discuss key contributions, directions for future research, and practical implications.

Table 8. Computation of Direct, Indirect, and Total Effects from Mediation Analysis

Sales meetings	Coefficient	Bootstrapped std. err.	$P > \|z\|$
Direct effect β_1	0.22	0.05	<0.001
Indirect effect $\beta_2\beta_3$	0.83	0.04	<0.001
Total effect $\beta_1 + \beta_2\beta_3$	1.04	0.02	<0.001

Notes. β_1 = path coefficient for *D3* effect on *sales meetings*. β_2 = path coefficient for *logins* effect on *sales meetings*. β_3 = path coefficient for *D3* effect on logins.

6.1. Contributions to the Literature

First, we contribute a human-centered approach to AI that shifts the focus from AI back to humans. By prioritizing human experts' information-processing needs and tailoring their interaction with AI accordingly, this approach enables humans to leverage their unique information-processing advantages when working with AI on unstructured tasks. *Human-centered AI* thus contrasts with recent work prioritizing the gradual automation of tasks through AI (Kesavan and Kushwaha 2020, Sun et al. 2022). While automation may be suitable for routine tasks, unstructured tasks require effective human-AI interaction. Our human-centered approach provides empirically validated insight into designing supportive contexts for human-AI interaction, extending prior research that primarily focused on the division of labor between humans and AI (Shrestha et al. 2019, Puranam 2021, Choudhary et al. 2023). This redirection is significant, as leveraging the information-processing

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complementarities between humans and AI forms the core rationale for human-AI interaction in augmentation research (Choudhury et al. 2020, Vomberg et al. 2023, Boyacı et al. 2024).

Second, our findings address discrepancies between theoretical assumptions and empirical observations regarding the augmentation-performance relationship (Kim et al. 2024, Hillebrand et al. 2025). We extend the literature by introducing a contingency perspective, demonstrating that while information-processing complementarities of humans and AI are a necessary criterion (Murray et al. 2021, Raisch and Krakowski 2021), they are not sufficient for performance benefits to materialize. Specifically, we show that tailoring human-AI interaction to individuals' cognitive needs enables performance gains, whereas untailored augmentation can even lead to negative performance outcomes.

We also contribute to the understanding of learning effects in human-AI interaction (Gaessler and Piezunka 2023, Te'eni et al. 2023, Raisch and Fomina 2025), demonstrating a direct relationship between increased AI utilization (D3 versus D2) and subsequent performance gains. These benefits extend beyond AI's inherent autonomous improvement through machine learning and human-AI complementarities. Furthermore, our findings enrich the ongoing debate on algorithm aversion (Dietvorst et al. 2018, Liu et al. 2023) versus algorithm appreciation (Logg et al. 2019, Fügener et al. 2022). The observed variation in AI utilization appears to reflect human learning and rational behavior rather than systematic bias (see also de Véricourt and Gurkan 2023).

6.2. Future Research

Our human-centered approach to AI focuses on tailoring human-AI interaction contexts to individuals' cognitive styles, but other adjustments, such as tailoring these contexts to individuals' experience (Wang et al. 2024), may also be valuable. Future research could also explore the cost of tailoring, which was not addressed in this study. It would also be relevant to explore how our findings apply to newer AI technologies, such as generative AI, including large language models (Krakowski 2025, Townsend et al. 2025). While our insights focus on unstructured managerial tasks requiring sustained human involvement, future research could investigate the generalizability of this approach to functional domains beyond sales, industries outside pharmaceuticals, and broader collective and societal perspectives of human-centered AI (Capel and Brereton 2023). The reduced AI system utilization in D2 might partially result from within-group spillovers, suggesting a need for future research to re-evaluate these treatments at different levels of analysis or by using alternative randomization methods. Our results are power-constrained and should therefore be interpreted with caution. While such limitations are common in field settings (Frick et al. 2023), we prioritized studying a real-world setting with high ecological validity for exploring the emerging topic of human-AI interaction in unstructured managerial tasks. Nonetheless, we encourage future research to validate our findings in larger laboratory or field studies.

6.3. Practical Implications

AI experts and practitioners increasingly advocate for *human-centered AI* that more directly supports humans in their interactions with AI (Shneiderman 2022). A key aspect of this approach is ensuring that humans remain actively involved in AI applications to assume responsibility, prevent harm, and ensure societal benefits by selectively invoking, validating, or overriding AI outputs. Our findings highlight the importance of tailoring human-AI interaction contexts to cognitive styles as a critical factor in enabling human-centered AI. This study provides concrete insight into the design and implementation of human-centered AI, demonstrating how tailored human-AI interaction can significantly enhance performance in unstructured managerial tasks. These insights are essential since most managerial tasks are unstructured and not amenable to consistent or optimization-oriented automation (Raisch and Krakowski 2021).

In addition, our findings demonstrate that in practical business contexts, the implementation of an advanced AI system can yield inferior outcomes even when compared with a technologically inferior IT system (D1). Our combined quantitative and qualitative insights suggest that such systems cannot be evaluated solely through direct comparison but must also be assessed in terms of the disruptions they introduce to established workflows. In our study, untailored AI implementation, despite the AI system's technical superiority relative to the prior IT system, undermined existing work procedures, resulting in overall negative performance. This emphasizes the need for managers to adopt a comprehensive, human-centered perspective when implementing AI systems, considering their broader impact on employee workflows and organizational dynamics as they are gradually integrated into decision-making processes and assume tasks traditionally requiring human expertise.

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Endnotes

[1] Unstructured tasks refer to situations "that have not been encountered in quite the same form and for which no predetermined and explicit set of ordered responses exist" (Mintzberg et al. 1976, p. 246). In such situations, the limits of automation are especially evident, requiring a joint effort of humans and AI (Raisch and Krakowski 2021).

[2] Interaction refers to the relational exchange between humans and algorithms conducting work in organizations (Tarafdar et al. 2023).

[3] The conceptualization and measurement of cognitive styles have been debated widely in the management and psychology literatures (Allinson and Hayes 1996, Armstrong et al. 2012, Li and Armstrong 2015). The KAI psychometric concept is however "widely accepted" (Scott and Bruce 1994, p. 590) and "used extensively" (Cools and Van Den Broeck 2007, p. 367). There is strong support for the KAI's construct validity, which has proven stable over time and across contexts, with accumulated evidence confirming its reliability and generalizability (Clapp 1993, Bagozzi and Foxall 1995). The KAI further exhibits convergent validity (i.e., high correlation with related measures such as peer and managerial evaluations and alternative self-assessments) and discriminant validity (i.e., largely uncorrelated with distinct constructs such as cognitive ability, affect, and creativity) (Keller and Holland 1978, Hayes and Allinson 1994).

[4] Two participating sales experts left the company approximately one year after the study commenced. They were excluded from the analysis.

[5] We do not differentiate between repeated meetings or new client meetings. According to senior sales staff from the company, the industry is highly stable with an overwhelming majority of sales meetings being conducted with established clients and relationships.

[6] Given that the *units sold* exhibits skewness, we explored variable transformations and conducted a supplementary analysis (available in Online Appendix C) to validate the robustness of our findings.

[7] The calibration of the four parameters in D2 was recommended by senior management at the pharmaceutical company. At the outset of the experiment, they expected this setting to be optimal for governing sales expert interaction with the AI system.

[8] To validate our results, we conduct supplementary analyses using the two alternative *experience* measures, namely pharmaceutical sales experience and experience from working at this specific pharmaceutical company (see Table I1, (a) and (b) in Online Appendix I) and substituting *age* for *experience* (see Table I2 in Online Appendix I), with both alternative specifications leaving our results materially unchanged.

[9] In D2, the insight that a misaligned "work procedure" parameter outweighs the alignment of the three remaining parameters for adaptors also applies to innovators. For them, the aligned "work procedure" parameter effectively compensates for the adverse effects of the three misaligned parameters, as evidenced by small and inconsistent performances effects (i.e., a slight increase in the *sales meetings* by 0.12 and only a minor decrease in *units sold* by −75.75).

[10] Notably, the innovators in D2 also perform better than the control group in terms of *sales meetings*. This finding suggests that the non-tailored implementation with a flexible process already enables innovators to benefit from the AI system. However, the innovators' performance in D3 with the tailored context increases more strongly as compared to D2. We refer to Online Appendix P for a supplementary analysis of the login data in relation to potential learning curve patterns.

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